Major Bleeding and Thromboembolic Events in Veno-Venous Extracorporeal Membrane Oxygenation-Patients With Isolated Respiratory Failure

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Bleeding and thromboembolic events are common during veno-venous extracorporeal membrane oxygenation (vvECMO). It is unknown whether these complications are driven by the ECMO system itself, multiorgan-failure, or both. The aim of this study was to assess the prevalence of bleeding and thromboembolic events in patients with isolated respiratory failure. Patients with vvECMO were retrospectively included from March 2009 to October 2017. Exclusion included any organ failure other than respiratory. Major bleeding was defined as a decrease in hemoglobin ≥2 g/dl per 24 hours, the requirement for transfusion of ≥2 packed red blood cell concentrates per 24 hours, any retroperitoneal, pulmonary, central nervous system bleeding, or bleeding requiring surgery. Thromboembolic events were assessed by duplex sonography or CT scan. Of 601 patients, 123 patients with a mean age of 49 ± 15 years and a median Sepsis-related Organ Failure Assessment score of 8 (7–9) were eligible for the analysis. Major bleeding was observed in 73%; 35% of all bleedings occurred on the day of or after ECMO initiation. A more pronounced decrease of $PaCO_2$ after ECMO initiation was seen in patients with intracranial bleeding (ICB) compared with those without. Thromboembolic events were noted in 30%. The levels of activated prothrombin time, fibrinogen, platelet count, or D-dimers affected neither bleeding nor the prevalence of thromboembolic events. *ASAIO Journal* 2022; 68;1529–1535

Key Words: extracorporeal membrane oxygenation, ARDS, ECMO, bleeding, thromboembolic, complication

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The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

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Veno-venous extracorporeal membrane oxygenation (vvECMO) is of growing relevance in the treatment of refractory severe acute respiratory distress syndrome.[1] This rescue therapy allows to reduce potential harmful lung injury due to invasive mechanical ventilation by applying an ultraprotective ventilation strategy and at the same time allows eventually sufficient tissue oxygenation despite hypoxemia due to lung injury.[2] Recent studies, such as the extracorporeal membrane oxygenation to rescue acute lung injury in severe acute respiratory distress syndrome trial[3] or the efficacy and economic assessment of conventional ventilatory support *versus* extracorporeal membrane oxygenation for severe adult respiratory failure (CESAR)[4] trial demonstrated beneficial outcomes in patients with severe acute respiratory distress syndrome when treated with vvECMO compared with controls.

In critically ill patients, complication such as bleeding and thrombotic events arise frequently. Catheter associated thrombosis was reported in critical ill patients without ECMO in up to 33%.[5] In contrast to that, prevalences of major bleeding and thrombotic events in patients requiring vvECMO are reported in up to 79% and 85%, respectively. Both can contribute to increased morbidity and mortality.[6]

Patients in need of ECMO are in general the sickest patients in intensive care unit.[7] Despite applying vvECMO for respiratory failure, most of them have additional organ failure that may account for coagulation complications. However, it is unclear whether these excessive complication rates are driven by the ECMO system itself (*e.g.,* blood trauma[8]), by the more extensive illness of the patients or a combination of multiple factors.

Therefore, the aim of the study was to assess bleeding and thromboembolic events in patients requiring vvECMO without any other organ failure than respiratory and to identify potential predisposing factors of routinely acquired coagulation parameters.

Material and Methods

Study Population

This analysis included all adult patients (≥18 years) requiring vvECMO for severe respiratory failure ($PaO_2/FiO_2 < 85$ mmHg or refractory respiratory acidosis with pH < 7.25 on optimized positive end-expiratory pressure, usually ≥15 mm Hg) from March 2009 to October 2017 at the University Hospital Regensburg, Germany. All patients were cannulated percutaneously.

Patient data have been derived from the Regensburg ECMO-Registry. The inclusion period for the analysis was defined due to the introduction of a new electronic patient data management system in March 2009 and a change of anticoagulation strategy at our center from September 2017 onward. Exclusion criteria for this analysis predisposing for bleeding and thromboembolic were defined as:

- Hepatic impairment (defined as elevation of serum aspartat aminotransferase or alanine aminotransferase >10 times of upper reference value)[9]
- Kidney-failure (defined as requirement of renal replacement therapy or elevation of serum-creatinine >1.5 mg/dl following the Kidney Disease: Improving Global Outcome criteria)[10]
- Patients suffering from polytrauma
- Heparin-induced thrombocytopenia
- Pulmonary bleeding as reason for respiratory failure requiring vvECMO
- Underlying diseases with presumable effects on coagulation, for example, hematologic disorder causing thrombocytopenia

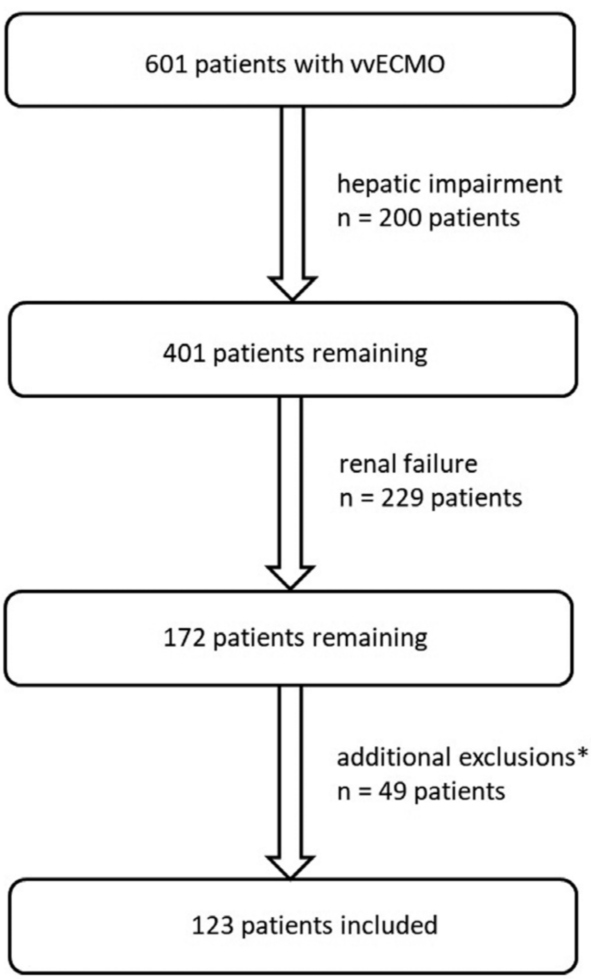

Figure 1. Flowchart of the retrospective observational study of the prospective ECMO registry Regensburg. *Additional exclusions: 25 patients with polytrauma, 13 patients with a documented diagnosis of either renal or hepatic failure, which were not excluded via the prior algorithm, five patients with underlying hematologic disorder causing thrombocytopenia, two patients with pulmonary bleeding as reason for vvECMO, two patients with HIT, one patient with drug-related thrombocytopenia, one patient with "acquired clotting factory deficiency" before vvECMO. ECMO, extracorporeal membrane oxygenation; vvECMO, venovenous ECMO.

After exclusion, 123 patients were eligible for the analysis (Figure 1).

The requirement of individual patient consent and necessity of approval for the data report were waived by the local ethics committee because of the study's design and data collection from routine care. The study was conducted according to the Declaration of Helsinki on Good Clinical Practice.

Anticoagulation Strategy

Anticoagulation was conducted as follows: At the time of cannulation, a bolus of up to 5,000 international units (IU) of unfractionated heparin according to provider's recommendation (80 IU/kg) was administered. After that, unfractionated heparin was continuously intravenously infused according to provider's recommendation (18 IU/kg/h). The flow rate was titrated and adapted to the activated prothrombin time (aPTT) target range of 50 ± 5 seconds according to former recommendations.[11,12] aPTT was checked every 4 to 6 hours until target range was achieved. More details are provided in a previously published report.[13]

Bleeding Complications

Major bleeding events were defined following the recommendations of the Extracorporeal Life Support Organization anticoagulation guidelines[14] as a decrease in hemoglobin level ≥2 g/dl within 24 hours, any retroperitoneal, pulmonary, or central nervous system bleeding or bleeding requiring surgical intervention. According to Schulman *et al.*[15] requirement of red blood cell transfusion has been simplified to two or more packed red blood cell concentrates from the European Life Support Organization weight-based definition. Bleeding was evaluated daily.

Thromboembolic Complications

Thromboembolic events were assessed by duplex sonography within 3 days after decannulation or CT scan as previously published.[13] Duplex sonography was performed by specially trained physicians. Diagnosis was based on incompressibility of the vein, absence of flow, and presence of thrombus.[16] The jugular vein was examined down to the insertion of the subclavian vein, and the femoral cannulation site was examined from the popliteal vein to the common iliac vein. Thrombosis was classified as relevant if the lumen was obstructed by >50%, since a larger thrombosis is a risk factor for thrombosis extension and bears a higher risk for pulmonary embolism.[17] Additionally, all available CT scans, which were made for various clinical indications during the hospital stay, were analyzed for pulmonary embolism and thrombosis.[13]

Laboratory Chemistry

Laboratory parameters such as aPTT, D-dimers, Quick, fibrinogen level, and platelet count were assessed daily. The Quick-test is routinely used in Germany and measures the extrinsic system of the coagulation pathway; thus, this test is comparable with international normalized ratio. The following laboratory methods have been used: "Pathromtin SL" for aPTT, "Thromborel S" for Quick, "Multifibren U" for fibrinogen, "Innovance D-Dimer" for D-dimers. All tests were provided by the company Siemens, Munich, Germany, and the analyzing system was the Siemens BCS XP.

Statistical Analysis

Nominal data were analyzed using χ^2 test. Quantitative data were tested for normal distribution using Kolmogorov-Smirnov test. Normally distributed quantitative variables were analyzed using students t-test. Non-normally distributed quantitative data are expressed as median (25th–75th percentile) and were compared using Mann-Whitney *U* test. All reported *p* values were two-sided and a *p* value ≤0.05 was considered statistically significant. Data entry and calculation were done using Microsoft EXCEL365 ProPlus (Microsoft, USA) and IBM SPSS Statistic software version 25.0 (SPSS Inc. Chicago, IL, USA).

Results

Study Population

From March 2009 to October 2017, a total of 601 patients were treated with vvECMO in our center. Four hundred seventy-eight (80%) had at least one additional organ failure and therefore 123 (20%) patients with isolated respiratory failure met the inclusion criteria (Figure 1). The mean age and days on ECMO were similar in both groups (isolated respiratory failure *versus* entirety of patients: 49 ± 15 *vs.* 50 ± 17; $p = 0.252$; 8 [5–13] *vs.* 9 [5–14]; $p = 0.769$, respectively). Eighty-six (70%) patients were cannulated with two single-lumen cannula and 37 (30%) patients with a double-lumen cannula.

Compared with the entirety of patients, the isolated respiratory failure cohort had a lower body mass index (26.2 ± 7.2 *vs.* 27.8 ± 8.67 kg/m^2; $p = 0.005$), consisted of fewer male patients (55% *vs.* 68%; $p = 0.003$) and had a Lower Sepsis-related Organ Failure Assessment (SOFA) score at baseline (8 [7–9] *vs.* 11 [8–15]; $p < 0.001$). Overall mortality was lower in the group with isolated respiratory failure compared with those with additional organ failure (18% *vs.* 39%; $p < 0.001$).

Major Bleedings

Ninety patients (73%) had at least one major bleeding event. In total, 196 major bleedings occurred resulting in two or more bleeding episodes per patient. Within the first 2 days of ECMO therapy 35% of all major bleedings occurred. Eight patients (9%) suffered from intracranial hemorrhage and 13 patients (14%) underwent surgery or major intervention because of bleeding—a detailed analysis of the interventions is listed in Table (Supplemental Digital Content 1, http://links.lww.com/ASAIO/A795). Fifty-five patients (61%) required a transfusion of ≥2PRBC/24 hours. Eighty-nine patients (99%) had documented Hb-loss of ≥2 g/dl/24 hours at least once during vvECMO. An allocation of all major bleeding events across the used definition is reported in Table (Supplemental Digital Content 2, http://links.lww.com/ASAIO/A795). Patients with major bleeding had a higher SOFA score, stayed longer on ECMO, and had a higher mortality compared with those without major bleeding (Table 1). In a multivariate model adjusting for demographics (age, sex, body mass index) and severity of critical illness (SOFA), higher SOFA scores were associated with major bleeding events (B [95% confidence intervals]: 1.446 [1.168–1.840]; $p = 0.001$). Cannulation technique did not affect the major bleeding rate (single-lumen *vs.* double-lumen: 65 [70%] *vs.* 25 [68%] patients; $p = 0.628$). The most frequent underlying disease leading to ECMO was pneumonia (79% of all cases). No significant differences between occurrence of major bleeding and the diagnosis group were seen ($p = 0.171$; Table, Supplemental Digital Content 3, http://links.lww.com/ASAIO/A795). Presence of major bleeding was not associated with significantly higher median levels in common inflammation parameters as C-reactive protein (CRP) (Major bleeding 118 mg/dl *vs.* no major bleeding 112 mg/dl; $p = 0.956$) or leukocyte count (major bleeding 12.0/nl *vs.* no major bleeding 11.9/nl; $p = 0.725$). Patients with major bleeding had a similar rate of thromboembolic events compared with those without and vice versa ($p = 0.610$; Table, Supplemental Digital Content 4, http://links.lww.com/ASAIO/A795).

Table 1. Outcome Comparison Regarding Major Bleeding

	Major Bleeding, n = 90	No Major Bleeding, n = 33	*p*
Death during ECMO	15 (17%)	1 (3%)	0.046
Hospital mortality	20 (22%)	2 (6%)	0.038
Days on ECMO	9 (6–14)	6 (5–8)	0.011
SOFA score	8 (7–10)	7 (6–8)	0.001

Data are expressed as n (%), median (25th percentile–75th percentile).

ECMO, extracorporeal membrane oxygenation; SOFA, sepsis-related organ failure assessment.

Laboratory Parameters With Respect to Major Bleeding

The median level of aPTT, fibrinogen, and D-dimers did not differ significantly comparing patients with major bleeding to those without (aPTT: 48 sec [42–54] *vs.* 49 sec [46–55]; $p = 0.389$; fibrinogen: 433 mg/dl [319–559] *vs.* 464 mg/dl [379–534]; $p = 0.644$; D-dimers: 10 mg/l [4–18] *vs.* 7 mg/l [3–15]; $p = 0.141$; Figure 2A–C). To rule out mutual interdependencies between aPTT and bleeding, we also tested the *median aPTT until first major bleeding event*. No significant difference was seen (46 sec [39–58] *vs.* 52 sec [47–61]; $p = 0.13$). The heparin dosage did not differ significantly in patients with major bleeding compared to those without (Table, Supplemental Digital Content 5, http://links.lww.com/ASAIO/A795).

The median platelet count during the ECMO run was significantly lower in patients with major bleeding (156/nl [120–215] *vs.* 198/nl [145–287]; $p = 0.02$; Figure 3A), whereas the median platelet count until first major bleeding was not different between groups (194/nl [148–269] *vs.* 198/nl [145–287]; $p = 0.975$; Figure 3B).

As stated earlier, 35% of all bleedings occurred within 2 days after cannulation. Neither median aPTT, nor Quick, platelet count, fibrinogen level, or D-dimers before implantation of vvECMO, 2 hours after implantation or on the second day of ECMO differed significantly in between groups regarding occurrence of major bleeding on the first 2 days of ECMO.

Arterial Carbon Dioxide Partial Pressure and Intracranial Bleeding

All ICB events occurred within 4 days after initiation of ECMO. There was no significant difference between patients with and without ICB accounting for median platelet count,

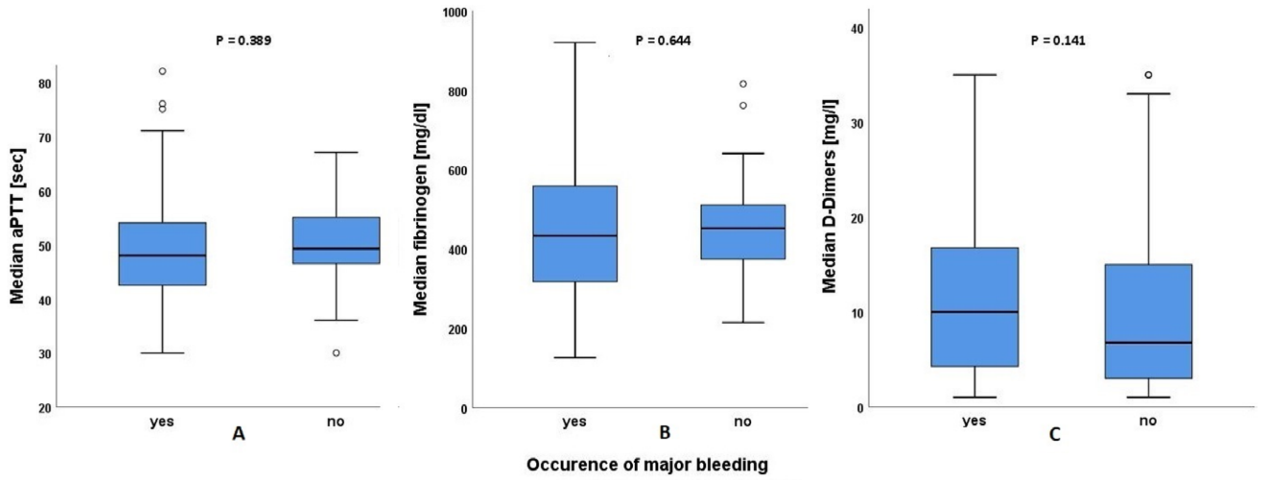

Figure 2. Major bleeding and laboratory results. full color online

Quick, aPTT, or fibrinogen level. However, patients with ICB had a significantly higher median $PaCO_2$ before cannulation (90 mmHg [70–125] *vs.* 67 mmHg [53–83]; $p = 0.027$; Figure 4A). Moreover, the difference of $PaCO_2$ before and after starting ECMO ($\Delta PaCO_2$: 50 mmHg [34–84] *vs.* 28 mmHg [18–44]; $p = 0.044$; Figure 4B) was higher in those suffering from ICB.

Thromboembolic Events

Of 123 included patients, 113 (92%) have been evaluated for the occurrence of thromboembolic events as defined. Out of those, 34 patients (30%) had documented pulmonary embolism or venous thrombosis with occlusion of >50% of the luminal diameter. Twenty-four (70%) patients had only thrombosis, four (12%) had only pulmonary embolism, six (18%) had both. Those with a thromboembolic event had a significantly lower mortality during vvECMO (0% *vs.* 18%; $p = 0.008$) and lower all-cause mortality (3% *vs.* 24%; $p = 0.008$) compared with those without. No significant effects were seen for time on ECMO or SOFA score. Neither median overall Quick, nor aPTT, platelet count, fibrinogen level, or D-dimers showed significant differences comparing the groups with and without thromboembolic events (Table 2). The heparin dosage did not differ significantly in patients with a thromboembolic event compared to those without (Table, Supplemental Digital Content 5, http://links.lww.com/ASAIO/A795). Cannulation technique did not affect the rate of thromboembolic events (single-lumen *vs.* double-lumen: 25 [29%] *vs.* 9 [24%] patients; $p = 0.934$). The time on ECMO in days was not associated with the occurrence of a thromboembolic event (B [95% confidence intervals]: 0.993 [0.954–1.033]; $p = 0.721$).

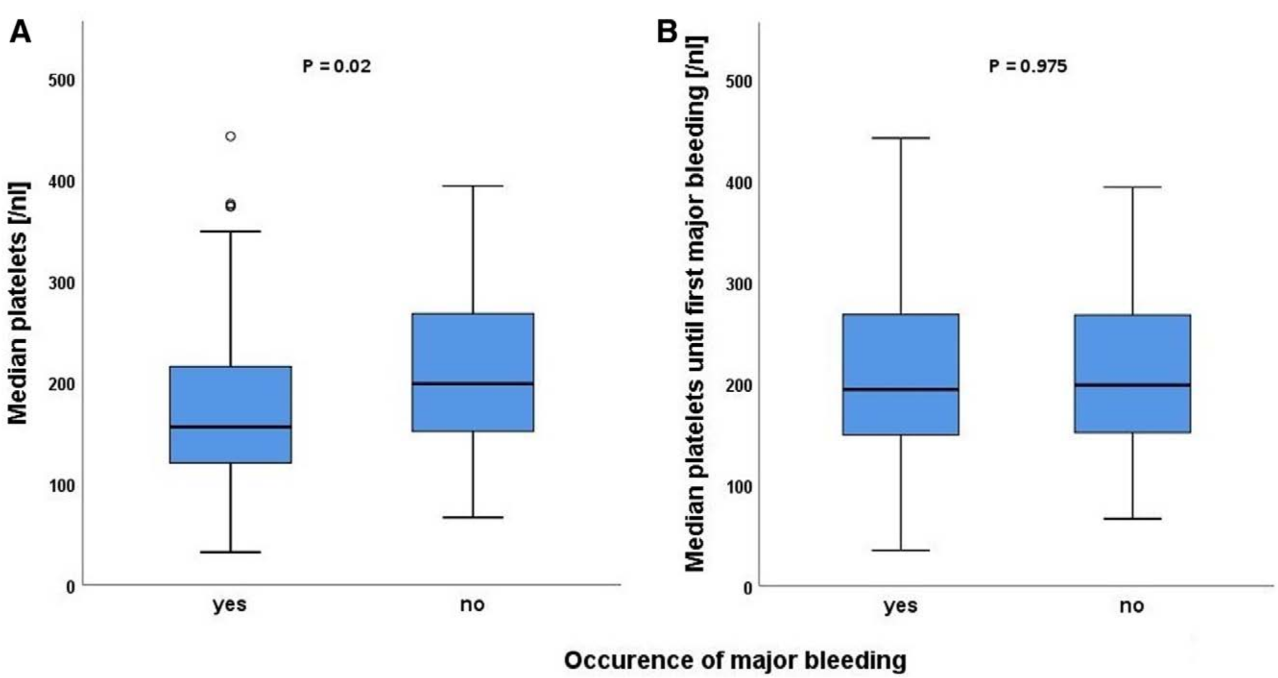

Figure 3. Major bleeding and platelets. full color online

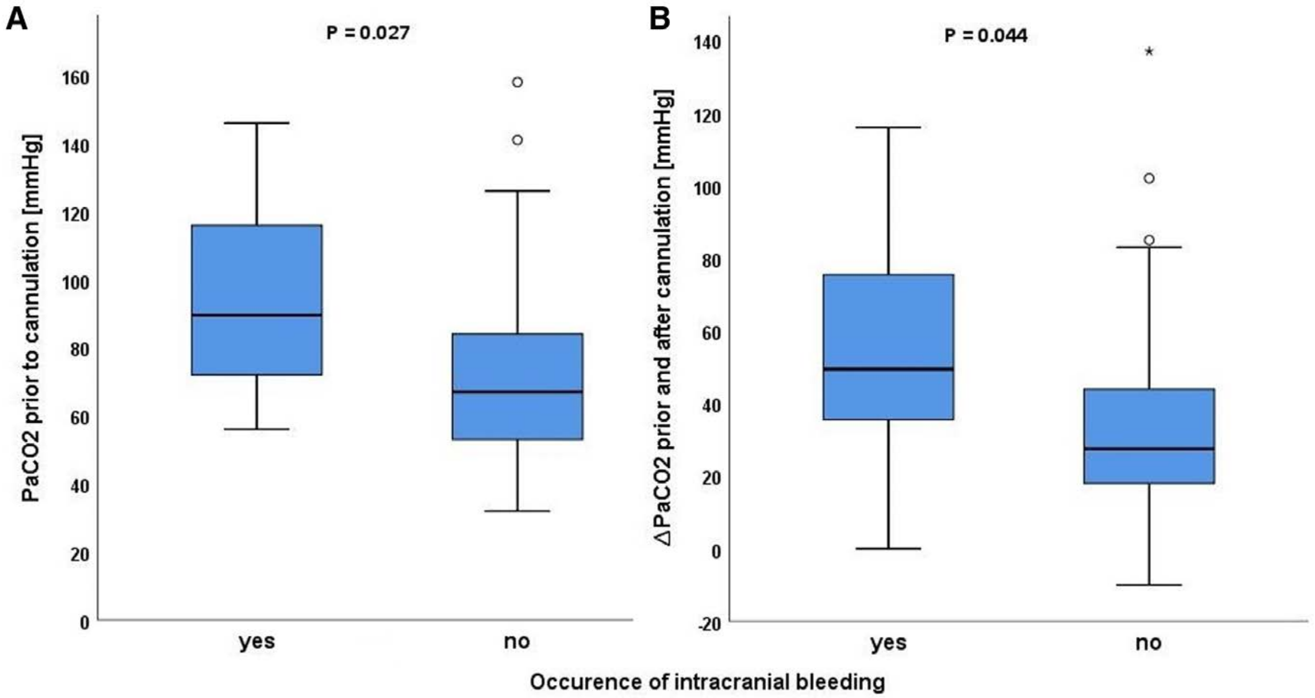

Figure 4. Intracranial bleeding and $PaCO_2$. full color online

Discussion

This study provides interesting insights in major bleeding and thromboembolic complications in patients treated with vvECMO for isolated respiratory failure. The strength of this study investigating a selected population of patients with single organ failure must be emphasized. To investigate bleeding complications of ECMO with less multifactorial influences is only possible in animal studies. Therefore, to our opinion, this patient population of isolated respiratory failure is the best way to focus on ECMO effects.

We observed a high rate of bleeding complications (73%) and a moderate rate of thromboembolic events (30%). Major bleeding was associated with a higher mortality but was not associated with significant differences in any of the analyzed coagulation parameters. The focus on plasma-based tests and lack of information about cellular interaction in our study might contribute to this result. It is debated in the literature, if the aPTT is the best monitoring parameter for anticoagulation with unfractionated heparin.[18,19] Others, like Anti-Xa-activity, might be advantageous. However, to date, the aPTT remains the widely accepted standard.[20–22] ICB was present in eight cases and associated with higher levels of $PaCO_2$ before cannulation and a higher difference of $PaCO_2$ before and 2 hours after starting ECMO. Thromboembolic events showed an inverse association with mortality rates and were not associated with significant differences regarding the coagulation parameters.

Bleeding Events

We identified a high number of bleeding events following the applied definition. Bleeding rates reported by others[6] vary between 0% and 91%. These differences might derive from different definitions of bleeding and our definition is rather liberal. A direct comparison with other studies[23,24] is not feasible because of the highly selected collective in this study. However, it is important to focus on bleeding in future studies, as in this

Table 2. Comparison of Patients With to Those Without Major Thromboembolic Events

	Thromboembolic Event, n = 34	No Thromboembolic Event, n = 89	*p*
Death during ECMO	0 (0%)	16 (18%)	0.008
Hospital mortality	1 (3%)	21 (24%)	0.008
Days on ECMO	8 (7–14)	7 (5–13)	0.951
SOFA score	8 (7–8)	8 (3)	0.688
Laboratory parameters			
Quick, %	80 (72–92)	82 (705–92)	n.s.
aPTT, sec	48 (45–54)	49 (44–55)	n.s.
Platelet count, /nl	164 (117–231)	164 (126–222)	n.s.
Fibrinogen, mg/dl	422 (376–561)	439 (329–556)	n.s.
D-dimers, mg/l	12 (4–27)	8 (4–16)	n.s. (0.072)

Data are expressed as n (%), median (25th–75th percentile).
aPTT, activated partial thromboplastin time; ECMO, extracorporeal membrane oxygenation; SOFA, sepsis-related organ failure assessment.

analysis patients with major bleeding required significantly longer ECMO support and had higher mortality rates. The latter has been reported similarly in some but not all studies in a review by Sklar *et al.*[6] The most frequent condition leading to ECMO was pneumonia, which itself is a proinflammatory state. Major bleeding was not associated with inflammatory parameters such as CRP or leukocytes, which suggests inflammation might not be the driving factor affecting hemostasis. Neither median Quick, aPTT nor fibrinogen level affected major bleeding events. However, the median platelet count was significantly lower in patients with at least one major bleeding.

Activated Partial Thromboplastin Time and Bleeding

A review in patients with vvECMO support identified an aPTT target >60 seconds as a risk factor for hemorrhagic complications.[6] In contrast, we found no significant differences in the median aPTT in patients with major bleeding events compared with those without. A possible explanation is our lower aPTT target of 50 ± 5 seconds, which might be below the threshold to be associated with bleeding.

A major confounding factor might be mutual interdependencies: In clinical routine bleeding may lead to reduced anticoagulation and thus result in normalization of the aPTT. To address this, we also analyzed the *median aPTT until the first major bleeding event*—this approach did not affect our findings.

Platelets and Bleeding

Both thrombocytopenia and impaired platelet function are common findings in ECMO patients.[25] In our study, major bleeding was associated with significantly lower platelet counts. However, whereas the difference in platelet counts is statistically significant, it cannot be classified as clinically relevant considering both were within the reference range. Platelet function might have been impaired,[26] which we did not assess. Furthermore, thrombocytopenia may be interpreted as a consequence of bleeding as well as its cause. To elucidate this *chicken or the egg causality dilemma*, we analyzed the median platelet count until the first major bleeding, which did not differ significantly in between the two groups.

Fibrinogen and Bleeding

In our study, major bleeding was not associated with lower fibrinogen levels. Device-induced hyperfibrinolysis is an infrequent but feared complication of ECMO, as it may lead to an exchange of the oxygenator or bleeding diathesis.[8] Our results may be explained by the rarity of the event itself and the relatively healthy study collective. Being an acute phase protein, the fibrinogen-dynamic might not reflect the coagulative situation but the inflammatory state. A comparison of the dynamic of the fibrinogen and CRP levels supports this interpretation (Figure, Supplemental Digital Content 1, http://links.lww.com/ASAIO/A795).

Intracranial Bleeding

ICB was present in 7% of all patients and 75% of ICBs were diagnosed within the first 2 days after cannulation. This rate is similar to other studies.[27–29] The most evident factors affecting bleeding in this critical period are the start of anticoagulation and the decrease of $PaCO_2$ before and after cannulation.[28] None of the measured coagulation parameters differed significantly before or 2 hours after the cannulation. Neither did their median levels within the first 2 days of ECMO. This contrasts the hypothesis that the beginning of anticoagulation is the leading factor of those early bleedings. In 2020 a large, retrospective cohort-study[30] showed that extensive relative decreases in $PaCO_2$ were independently associated with increased risk of neurologic complications. In our study, the $PaCO_2$ before cannulation and the absolute reduction of $PaCO_2$ pre *versus* 2 hours postcannulation were significantly higher in patients with an ICB. According to this observation, in cases with strongly elevated $PaCO_2$ levels before vvECMO-cannulation, a careful initial titration of sweep gas flow might be advisable. To accomplish this, we initiate ECMO treatment with 1.5–2 L/min blood flow and 1–3 L/min sweep flow. Further adaptions are made with respect to arterial blood gas analysis, drawn shortly after initiation of ECMO therapy. However, further investigation is required.

Thromboembolic Events

This study showed a prevalence of thromboembolism of 30% and higher mortality rates were associated with lower rates of thromboembolic events. This might be a consequence of routinely performing duplex sonography after decannulation and not regularly during the extracorporeal therapy, due to technical limitations. Consequently, the rate of thromboembolic events in deceased patients must be considered underreported and vice versa mortality analysis in patients with thromboembolic events is to be interpreted with caution. However, absolute numbers in those suffering from thromboembolism are comparable to previous studies in patients with multiple organ failure.[13] In contrast to others,[31] none of the analyzed coagulation-related laboratory parameters were significantly different in patients with and without thromboembolic events.

In our study, D-dimers were higher by trend in patients with thromboembolic events. This interesting finding needs further investigation as D-dimers have been identified as both an early marker for necessity of oxygenator exchange[32] as well as predictor for deep vein thrombosis after decannulation.[13]

Limitations

First, due to the highly selected collective conclusions cannot simply be transferred to other cohorts undergoing vvECMO in general. Second, bleeding was assessed retrospectively with its known limitations and in this setting effects of dilution cannot be excluded definitely. Third, routine duplex sonography was conducted after decannulation, thus affecting our rates of thromboembolism in those surviving and nonsurviving. Fourth, we cannot exclude that some details of management had changed over time; however, laboratory methods remained the same. Fifth, anti-Xa-activity has not been measured. Sixth, there is a focus on plasma-based tests and disregard of cellular interaction regarding the analyzed coagulation parameters. Finally, despite excluding many predisposing conditions for bleeding by the selection of the study cohort, we cannot exclude that other factors might have affected the bleeding events.

Conclusions

Bleeding and thromboembolic complications are frequent events in patients requiring vvECMO for isolated respiratory failure. Decline of $PaCO_2$ after ECMO initiation was higher in those suffering from ICB. Laboratory parameters were neither associated with bleeding nor with thromboembolic events in this specific population, and in clinical practice, all contributing factors together with the clinical presentation of the patient need to be taken into account.

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